Linear Regression and OLS

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Linear Regression & OLS Practice Exercises**

A biostatistician is investigating the relationship between the dosage of a new experimental drug (measured in milligrams) and the reduction in systolic blood pressure (in mmHg) for patients with mild hypertension. They collected data from $n = 200$ patients and fit a simple linear regression model. The model's output yielded an estimated intercept of $\hat{\beta}_0 = 5.2$ and an estimated slope of $\hat{\beta}_1 = -0.8$. The standard error of the estimated slope is $\text{SE}(\hat{\beta}_1) = 0.15$. A 95% confidence interval for the true slope β_1 was calculated to be $[-1.12, -0.48]$. The biostatistician then constructs a hypothesis test to determine if the true slope is significantly different from zero. To do this, they calculate a test statistic. If the same α level used for the confidence interval were applied to a two-tailed hypothesis test of $H_0 : \beta_1 = 0$ versus $H_a : \beta_1 eq 0$, what would be the approximate p-value associated with this test statistic? Show your complete algebraic work.

Final Answer: Less than 0.05. The problem asks for the approximate p-value of a two-tailed hypothesis test for the slope parameter β_1 in a simple linear regression model. We are given a 95% confidence interval for β_1 as $[-1.12, -0.48]$. A confidence interval and a hypothesis test are related. For a $100(1-\alpha)\%$ confidence interval, if the null hypothesis value (in this case, $\beta_1 = 0$) is *outside* the confidence interval, we reject H_0 at the α significance level. Conversely, if th.

2 **[Foundational]** • **Target: Simple Linear Regression and Residuals**

Consider a simple linear regression model where the predicted value $\hat{y}$ is given by $\hat{y} = b_0 + b_1 x$. If we have two data points $(x_1, y_1) = (2, 5)$ and $(x_2, y_2) = (4, 9)$, and the Ordinary Least Squares (OLS) method yields the regression line $\hat{y} = 1 + 2x$, what is the predicted value $\hat{y}$ for $x = 3$? Show your complete algebraic work.

Final Answer: 7. The problem provides a simple linear regression equation $\hat{y} = b_0 + b_1 x$ and specific values for the intercept $b_0 = 1$ and the slope $b_1 = 2$. We are asked to find the predicted value $\hat{y}$ for a given value of x. We substitute $x = 3$ into the regression equation: $\hat{y} = 1 + 2(3)$ $\hat{y} = 1 + 6$ $\hat{y} = 7$ Therefore, the predicted value $\hat{y}$ for $x = 3$ is 7. The data points $(2, 5)$ and $(4, 9)$ are consistent with this regression line: for $x = 2$.

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3 [Foundational] • Target: Analytical OLS Formulas and Covariance

Consider a simple linear regression model where the predicted value $\hat{y}$ is related to the independent variable x by the equation $\hat{y} = mx + c$. If the line of best fit passes through the points $(1, 5)$ and $(3, 11)$, what is the slope m of this line? Show your complete algebraic work.

Final Answer: 3. The slope m of a line passing through two points (x_1, y_1) and (x_2, y_2) is given by the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Given the points $(1, 5)$ and $(3, 11)$, we have $x_1 = 1$, $y_1 = 5$, $x_2 = 3$, and $y_2 = 11$. Plugging these values into the formula, we get $m = \frac{11 - 5}{3 - 1} = \frac{6}{2} = 3$. Therefore, the slope of the line of best fit is 3.

4 [Foundational] • Target: Matrix Normal Equations and Projections

A fitted simple linear regression model is $\hat{y} = 7x + 18$. What is the predicted target value $\hat{y}$ when the input feature is $x = 4$? Show your complete algebraic work.

Final Answer: 46. Substitute $x = 4$ into the regression equation: $\hat{y} = 7(4) + 18 = 28 + 18 = 46$.

5 [Foundational] • Target: Goodness of Fit and Regularization

A simple linear regression model has been fitted to a dataset. The model equation is given by $\hat{y} = ax + b$. The regression line passes through the points $(2, 10)$ and $(5, 22)$ in the feature-target space. Determine the value of the intercept b.

Final Answer: -2. The problem states that the linear regression line $\hat{y} = ax + b$ passes through two points: $(x_1, y_1) = (2, 10)$ and $(x_2, y_2) = (5, 22)$. We can use these two points to form a system of linear equations to solve for the unknown parameters a (the slope) and b (the intercept). Substituting the first point $(2, 10)$ into the equation $\hat{y} = ax + b$ gives: $10 = a(2) + b$ (Equation 1) Substituting the second point $(5, 22)$ into the equation $\hat{y} = ax + b$ gives: $22 = a(5) + b$.

6 [Foundational] • Target: Linear Regression & OLS Practice Exercises

Consider a simple logistic regression model where the decision boundary is defined by the equation $3x + 4y - 12 = 0$. This equation represents a linear boundary in a 2D feature space. What is the y-intercept of this decision boundary? Show your complete algebraic work.

Final Answer: $y = 3$. The equation of the decision boundary is given by $3x + 4y - 12 = 0$. To find the y-intercept, we set $x = 0$ and solve for y. Substituting $x = 0$ into the equation: $3(0) + 4y - 12 = 0$ $4y - 12 = 0$ $4y = 12$ $y = \frac{12}{4}$ $y = 3$ Therefore, the y-intercept of the decision boundary is 3.

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7 [Foundational] • Target: Simple Linear Regression and Residuals

Consider a dataset of n observations (x_i, y_i) for $i = 1, \ldots, n$, where x_i are scalar predictor variables and y_i are scalar response variables. The ordinary least squares (OLS) estimates for the intercept β_0 and the slope β_1 of the linear model $y = \beta_0 + \beta_1 x + \varepsilon$ are given by $\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$ and $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$. Suppose we have a dataset where $n = 5$, $\sum x_i = 15$, $\sum y_i = 30$, $\sum x_i^2 = 55$, $\sum y_i^2 = 210$, and $\sum x_i y_i = 105$. A new data point (x_6, y_6) is added to the dataset. If the new regression line that minimizes the sum of squared errors for the $n + 1 = 6$ points is parallel to the original regression line (i.e., $\hat{\beta}_1^{\text{new}} = \hat{\beta}_1^{\text{old}}$), what is the value of y_6 in terms of x_6? Show your complete algebraic work.

Final Answer: $y_6 = 1.5x_6 + 1.5$. First, we calculate the means: $\bar{x}^{\text{old}} = \frac{\sum x_i}{n} = \frac{15}{5} = 3$ and $\bar{y}^{\text{old}} = \frac{\sum y_i}{n} = \frac{30}{5} = 6$. Next, we calculate the sums of squares and cross-products for the original dataset: $S_{xx} = \sum (x_i - \bar{x})^2 = \sum x_i^2 - n\bar{x}^2 = 55 - 5(3^2) = 55 - 45 = 10$ $S_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y}) = \sum x_i y_i - n\bar{x}\bar{y} = 105 - 5(3)(6) = 105 - 90 = 15$ The original slope estimate i.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Analytical OLS Formulas and Covariance

Consider the matrix $A = (1, 1, 0; 1, 0, 1; 0, 1, 1)$. Let $\mathbf{b} = (1; 1; 1)$. We want to find the vector $\hat{\mathbf{x}}$ that minimizes $Ax - b$. This is equivalent to finding the projection of $\mathbf{b}$ onto the column space of A, denoted by $\text{Col}(A)$. The normal equations for this least squares problem are $A^T A\mathbf{x} = A^T\mathbf{b}$. The unique solution $\hat{\mathbf{x}}$ to the normal equations yields the best approximate solution. Calculate the value of $\hat{\mathbf{x}}_1$, the first component of $\hat{\mathbf{x}}$.

Final Answer: 1/2. The normal equations are given by $A^T A\mathbf{x} = A^T\mathbf{b}$. First, we compute $A^T A$ and $A^T\mathbf{b}$. \\ $A^T = (1, 1, 0; 1, 0, 1; 0, 1, 1)$ \\ $A^T A = (1, 1, 0; 1, 0, 1; 0, 1, 1)(1, 1, 0; 1, 0, 1; 0, 1, 1) = (2, 1, 1; 1, 2, 1; 1, 1, 2)$ \\ $A^T\mathbf{b} = (1, 1, 0; 1, 0, 1; 0, 1, 1)(1; 1; 1) = (2; 2; 2)$ \\ So, we need to solve the system.

9 [Intermediate] • Target: Matrix Normal Equations and Projections

Consider a 2D dataset represented by the points $A = (1, 2)$, $B = (4, 2)$, and $C = (2, 5)$. If this dataset undergoes a linear transformation represented by the matrix $M = (2, 0; 0, 1)$, what are the coordinates of the transformed points A', B', and C'? Show your complete algebraic work.

Final Answer: $A' = (2, 2), B' = (8, 2), C' = (4, 5)$. The transformation of a point $\mathbf{x}$ by a matrix M is given by $M\mathbf{x}$. We apply the matrix $M = (2, 0; 0, 1)$ to each point: For point $A = (1, 2)$, the transformed point A' is: $(2, 0; 0, 1)(1; 2) = (2 \times 1 + 0 \times 2; 0 \times 1 + 1 \times 2) = (2; 2)$ So, $A' = (2, 2)$. For point $B = (4, 2)$, the transformed point B' is.

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10 [Intermediate] • Target: Goodness of Fit and Regularization

Consider the following system of linear equations represented by the augmented matrix: $(1, 2, |, 5; 2, -1, |, 0)$. Use row reduction to transform the matrix into row-echelon form and determine the unique solution (x, y) for the system. What is the value of x? Show your complete algebraic work.

Final Answer: 1. We are given the augmented matrix: $(1, 2, |, 5; 2, -1, |, 0)$ To transform this matrix into row-echelon form, we can perform the following row operation: $R_2 \leftarrow R_2 - 2R_1$. $(1, 2, |, 5; 2 - 2(1), -1 - 2(2), |, 0 - 2(5))$ $(1, 2, |, 5; 0, -5, |, -10)$ Now, we can perform another row operation: $R_2 \leftarrow R_2/-5$. $(1, 2, |, 5; 0, 1, |, 2)$ This matrix is now in row-echelon form. The corresponding system of equations is: $x + 2y = 5$.

11 [Intermediate] • Target: Linear Regression & OLS Practice Exercises

Consider a simple logistic regression model where the decision boundary is defined by the linear equation $w_1x_1 + w_2x_2 + b = 0$. If the weights are $w_1 = 2$ and $w_2 = 3$, and the bias is $b = -7$, what is the equation of the decision boundary? Show your complete algebraic work.

Final Answer: $2x_1 + 3x_2 - 7 = 0$. The decision boundary in a logistic regression model is defined by the equation where the log-odds are zero. This corresponds to a linear combination of the input features and weights, plus the bias, being equal to zero. Given the weights $w_1 = 2$ and $w_2 = 3$, and the bias $b = -7$, the equation of the decision boundary is $w_1x_1 + w_2x_2 + b = 0$. Substituting the given values, we get $2x_1 + 3x_2 + (-7) = 0$, which simplifies to $2x_1 + 3x_2 - 7 = 0$.

12 [Intermediate] • Target: Simple Linear Regression and Residuals

Consider a simple linear regression model where a dependent variable y is predicted from an independent variable x. The model is given by the equation $y = \beta_0 + \beta_1x + \varepsilon$, where ε is the error term. Suppose we have estimated the regression line to be $\hat{y} = 3 + 2x$. If we observe an independent variable value of $x = 4$, what is the predicted value $\hat{y}$ for the dependent variable? Show your complete algebraic work.

Final Answer: 11. The problem asks for the predicted value $\hat{y}$ given an independent variable value $x = 4$ and a linear regression equation $\hat{y} = 3 + 2x$. We substitute the value of x into the equation: $\hat{y} = 3 + 2(4)$ $\hat{y} = 3 + 8$ $\hat{y} = 11$ Therefore, the predicted value for the dependent variable is 11. Common errors might involve incorrectly multiplying 2 by 4 or incorrectly adding 3 to 8.

13 [Intermediate] • Target: Analytical OLS Formulas and Covariance

Consider the following system of linear equations represented by an augmented matrix: $[1, 2, |, 5; 0, 1, |, 2]$. What is the value of x in the first equation? Show your complete algebraic work.

Final Answer: 3. The augmented matrix represents the system of equations: $x + 2y = 5$ and $y = 2$. We are asked to find the value of x. We can use substitution to solve for x. Substitute the value of y from the second equation into the first equation: $x + 2(2) = 5$. This simplifies to $x + 4 = 5$. Subtracting 4 from both sides gives $x = 1$. Therefore, the value of x in the first equation is 1.

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14 [Intermediate] • Target: Matrix Normal Equations and Projections

Consider the system of linear equations represented by the following augmented matrix: (1, 2, |, 5; 0, 1, |, 2) Use row reduction to determine the unique solution for x and y in the system. What is the value of x? Show your complete algebraic work.

Final Answer: 3. The augmented matrix represents the system of equations: 1) $x + 2y = 5$ 2) $y = 2$ Since the second equation directly gives the value of y as 2, we can substitute this into the first equation to find x. Substitute $y = 2$ into equation (1): $x + 2(2) = 5$ $x + 4 = 5$ $x = 5 - 4$ $x = 1$ Therefore, the unique solution is $x = 1$ and $y = 2$. The question asks for the value of x.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Goodness of Fit and Regularization • Cross-Lesson: 7.01 Logistic Regression and Classification

For a bivariate feature dataset, $\text{Cov}(x, y) = 36$, $\text{Var}(x) = 12$, $\bar{x} = 5$, and $\bar{y} = 35$. Compute the OLS regression slope $\hat{\beta}_1$ and intercept $\hat{\beta}_0$.

Final Answer: $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 20$. The OLS slope is $\hat{\beta}_1 = \frac{\text{Cov}(x,y)}{\text{Var}(x)} = \frac{36}{12} = 3$. The intercept is $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x} = 35 - (3)(5) = 35 - 15 = 20$.

16 [Challenging] • Target: Linear Regression & OLS Practice Exercises • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

For a bivariate feature dataset, $\text{Cov}(x, y) = 48$, $\text{Var}(x) = 16$, $\bar{x} = 5$, and $\bar{y} = 40$. Compute the OLS regression slope $\hat{\beta}_1$ and intercept $\hat{\beta}_0$.

Final Answer: $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 25$. The OLS slope is $\hat{\beta}_1 = \frac{\text{Cov}(x,y)}{\text{Var}(x)} = \frac{48}{16} = 3$. The intercept is $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x} = 40 - (3)(5) = 40 - 15 = 25$.

17 [Challenging] • Target: Simple Linear Regression and Residuals • Cross-Lesson: 1.01 Vectors and Embeddings in AI

For a bivariate feature dataset, $\text{Cov}(x, y) = 60$, $\text{Var}(x) = 20$, $\bar{x} = 5$, and $\bar{y} = 45$. Compute the OLS regression slope $\hat{\beta}_1$ and intercept $\hat{\beta}_0$.

Final Answer: $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 30$. The OLS slope is $\hat{\beta}_1 = \frac{\text{Cov}(x,y)}{\text{Var}(x)} = \frac{60}{20} = 3$. The intercept is $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x} = 45 - (3)(5) = 45 - 15 = 30$.

18 [Challenging] • Target: Analytical OLS Formulas and Covariance • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

For a bivariate feature dataset, $\text{Cov}(x, y) = 72$, $\text{Var}(x) = 24$, $\bar{x} = 5$, and $\bar{y} = 50$. Compute the OLS regression slope $\hat{\beta}_1$ and intercept $\hat{\beta}_0$.

Final Answer: $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 35$. The OLS slope is $\hat{\beta}_1 = \frac{\text{Cov}(x,y)}{\text{Var}(x)} = \frac{72}{24} = 3$. The intercept is $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x} = 50 - (3)(5) = 50 - 15 = 35$.

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19 [Challenging] • Target: Matrix Normal Equations and Projections • Cross-Lesson: 3.01 Gradient Descent and Optimization

For a bivariate feature dataset, $\text{Cov}(x, y) = 12$, $\text{Var}(x) = 4$, $\bar{x} = 5$, and $\bar{y} = 25$. Compute the OLS regression slope $\hat{\beta}_1$ and intercept $\hat{\beta}_0$.

Final Answer: $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 10$. The OLS slope is $\hat{\beta}_1 = \frac{\text{Cov}(x,y)}{\text{Var}(x)} = \frac{12}{4} = 3$. The intercept is $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 25 - (3)(5) = 25 - 15 = 10$.

20 [Challenging] • Target: Goodness of Fit and Regularization • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

For a bivariate feature dataset, $\text{Cov}(x, y) = 24$, $\text{Var}(x) = 8$, $\bar{x} = 5$, and $\bar{y} = 30$. Compute the OLS regression slope $\hat{\beta}_1$ and intercept $\hat{\beta}_0$.

Final Answer: $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 15$. The OLS slope is $\hat{\beta}_1 = \frac{\text{Cov}(x,y)}{\text{Var}(x)} = \frac{24}{8} = 3$. The intercept is $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 30 - (3)(5) = 30 - 15 = 15$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Linear Regression & OLS Practice Exercises

A fitted simple linear regression model is \(\hat{y} = 2x + 3 \). What is the predicted target value \(\hat{y} \) when the input feature is \(x = 4 \)?

a. **\(11 \) ✓**
b. \(14 \)
c. \(7 \)
d. \(8 \)

Substitute \(x = 4 \) into the regression equation: \(\hat{y} = 2(4) + 3 = 8 + 3 = 11 \).

22 [Foundational] • Target: Linear Regression & OLS Practice Exercises

A fitted simple linear regression model is \(\hat{y} = 3x + 6 \). What is the predicted target value \(\hat{y} \) when the input feature is \(x = 4 \)?

a. **\(18 \) ✓**
b. \(21 \)
c. \(14 \)
d. \(12 \)

Substitute \(x = 4 \) into the regression equation: \(\hat{y} = 3(4) + 6 = 12 + 6 = 18 \).

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23 [Foundational] • Target: Linear Regression & OLS Practice Exercises

A fitted simple linear regression model is $\hat{y} = 4x + 9$. What is the predicted target value $\hat{y}$ when the input feature is $x = 4$?

a. **25** ✓ **b.** 28

c. 21 **d.** 16

Substitute $x = 4$ into the regression equation: $\hat{y} = 4(4) + 9 = 16 + 9 = 25$.

24 [Foundational] • Target: Linear Regression & OLS Practice Exercises

A fitted simple linear regression model is $\hat{y} = 5x + 12$. What is the predicted target value $\hat{y}$ when the input feature is $x = 4$?

a. **32** ✓ **b.** 35

c. 28 **d.** 20

Substitute $x = 4$ into the regression equation: $\hat{y} = 5(4) + 12 = 20 + 12 = 32$.

25 [Foundational] • Target: Linear Regression & OLS Practice Exercises

A fitted simple linear regression model is $\hat{y} = 6x + 15$. What is the predicted target value $\hat{y}$ when the input feature is $x = 4$?

a. **39** ✓ **b.** 42

c. 35 **d.** 24

Substitute $x = 4$ into the regression equation: $\hat{y} = 6(4) + 15 = 24 + 15 = 39$.

26 [Foundational] • Target: Linear Regression & OLS Practice Exercises

A fitted simple linear regression model is $\hat{y} = 7x + 18$. What is the predicted target value $\hat{y}$ when the input feature is $x = 4$?

a. **46** ✓ **b.** 49

c. 42 **d.** 28

Substitute $x = 4$ into the regression equation: $\hat{y} = 7(4) + 18 = 28 + 18 = 46$.

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27 [Foundational] • Target: Linear Regression & OLS Practice Exercises

In a multiple linear regression model, the Sum of Squared Errors is \(\text{SSE} = 20 \) and the Total Sum of Squares is \(\text{SST} = 100 \). What is the coefficient of determination \(R^2 \)?

a. **\(R^2 = 0.80 \) ✓**

b. \(R^2 = 0.20 \)

c. \(R^2 = 0.90 \)

d. \(R^2 = 0.25 \)

The coefficient of determination measures the proportion of variance explained: \(R^2 = 1 - \frac{\text{SSE}}{\text{SST}} = 1 - \frac{20}{100} = 1 - 0.20 = 0.80 \).

28 [Foundational] • Target: Linear Regression & OLS Practice Exercises

In a multiple linear regression model, the Sum of Squared Errors is \(\text{SSE} = 40 \) and the Total Sum of Squares is \(\text{SST} = 200 \). What is the coefficient of determination \(R^2 \)?

a. **\(R^2 = 0.80 \) ✓**

b. \(R^2 = 0.20 \)

c. \(R^2 = 0.90 \)

d. \(R^2 = 0.25 \)

The coefficient of determination measures the proportion of variance explained: \(R^2 = 1 - \frac{\text{SSE}}{\text{SST}} = 1 - \frac{40}{200} = 1 - 0.20 = 0.80 \).

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Simple Linear Regression and Residuals

In simple linear regression $\hat{y} = 3.5x + 12$, what is the residual $e_i = y_i - \hat{y}_i$ for an observation at $(x_i, y_i) = (4, 30)$?

a. $+4$ ✓

b. -4

c. $+2$

d. -2

$\hat{y} = 3.5(4) + 12 = 14 + 12 = 26$. Residual $e = 30 - 26 = +4$.

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30 [Intermediate] • Target: Simple Linear Regression and Residuals

What is the mathematical condition guaranteed by the OLS fit regarding the sum of all residuals $\sum_{i=1}^{n} e_i$?

a. $\sum e_i = 0$ **exactly** ✓

b. $\sum e_i = 1$

c. $\sum e_i = n$

d. $\sum e_i = \text{SSE}$

The first normal equation $\frac{\partial \text{SSE}}{\partial \beta_0} = -2\sum(y_i - \beta_0 - \beta_1 x_i) = 0$ guarantees $\sum e_i = 0$.

31 [Intermediate] • Target: Simple Linear Regression and Residuals

Given $\bar{x} = 10$, $\bar{y} = 25$, and fitted slope $\hat{\beta}_1 = 2$, compute the y-intercept $\hat{\beta}_0$.

a. 5 ✓

b. 15

c. 20

d. 10

Regression line passes through the centroid: $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 25 - 2(10) = 5$.

32 [Intermediate] • Target: Analytical OLS Formulas and Covariance

For a dataset with $\text{Cov}(x, y) = 18$ and $\text{Var}(x) = 6$, compute the OLS slope $\hat{\beta}_1$.

a. 3.0 ✓

b. 0.33

c. 12

d. 108

$\hat{\beta}_1 = \frac{\text{Cov}(x,y)}{\text{Var}(x)} = \frac{18}{6} = 3.0$.

33 [Intermediate] • Target: Analytical OLS Formulas and Covariance

If the Pearson correlation coefficient between x and y is $r = 0.8$, with $\sigma_x = 2$ and $\sigma_y = 5$, find the slope $\hat{\beta}_1$.

a. 2.0 ✓

b. 1.25

c. 0.32

d. 4.0

$\hat{\beta}_1 = r\frac{\sigma_y}{\sigma_x} = 0.8\left(\frac{5}{2}\right) = 0.8(2.5) = 2.0$.

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34 [Intermediate] • Target: Analytical OLS Formulas and Covariance

If all x values in a dataset are scaled by a factor of 10, how does the OLS regression slope change?

a. **It is divided by 10 ✓**

b. It is multiplied by 10

c. It is multiplied by 100

d. It remains unchanged

$\beta_1' = \frac{\text{Cov}(10x, y)}{\text{Var}(10x)} = \frac{10\text{Cov}(x, y)}{100\text{Var}(x)} = \frac{1}{10}\beta_1$.

35 [Intermediate] • Target: Matrix Normal Equations and Projections

State the matrix Normal Equations for the OLS weight vector $\boldsymbol{\beta}$ given design matrix $\mathbf{X}$ and target $\mathbf{y}$.

a. $(\mathbf{X}^T\mathbf{X})\boldsymbol{\beta} = \mathbf{X}^T\mathbf{y}$ ✓

b. $\mathbf{X}\boldsymbol{\beta} = \mathbf{y}$

c. $(\mathbf{X}\mathbf{X}^T)\boldsymbol{\beta} = \mathbf{y}$

d. $\boldsymbol{\beta} = \mathbf{X}^T\mathbf{y}$

Minimizing $||\mathbf{y} - \mathbf{X}\boldsymbol{\beta}||^2 \Rightarrow -2\mathbf{X}^T(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = 0 \Rightarrow (\mathbf{X}^T\mathbf{X})\boldsymbol{\beta} = \mathbf{X}^T\mathbf{y}$.

36 [Intermediate] • Target: Matrix Normal Equations and Projections

When is the Gram matrix $(\mathbf{X}^T\mathbf{X})$ strictly invertible?

a. **When the columns of $\mathbf{X}$ are linearly independent (full column rank) ✓**

b. When $\mathbf{X}$ is a square matrix

c. When all elements in $\mathbf{X}$ are positive

d. When the target $\mathbf{y}$ has zero mean

Full column rank ensures $\mathbf{X}^T\mathbf{X}$ is positive definite and thus invertible.

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37 [Intermediate] • Target: Matrix Normal Equations and Projections

What geometric role does the hat matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$ play?

a. It orthogonal-projects target $\mathbf{y}$ onto the column space of $\mathbf{X}$ ($\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$) ✓

b. It inverts the covariance matrix

c. It computes feature cross-correlations

d. It normalizes features to zero mean

$\mathbf{H}$ satisfies $\mathbf{H}^2 = \mathbf{H}$ (idempotent) and $\mathbf{H}^T = \mathbf{H}$ (symmetric), the definition of orthogonal projection.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Goodness of Fit and Regularization • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

In an evaluation, $\text{SSE} = 25$ and $\text{SST} = 100$. Compute the coefficient of determination R^2.

a. 0.75 ✓

b. 0.25

c. 0.50

d. 0.80

$R^2 = 1 - \frac{\text{SSE}}{\text{SST}} = 1 - \frac{25}{100} = 0.75$.

39 [Challenging] • Target: Goodness of Fit and Regularization • Cross-Lesson: 3.01 Gradient Descent and Optimization

How does Ridge regression (L_2 penalty) resolve multicollinearity when $\mathbf{X}^T\mathbf{X}$ is ill-conditioned?

a. It adds $\lambda\mathbf{I}$ to $\mathbf{X}^T\mathbf{X}$, guaranteeing invertibility: $(\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}$ ✓

b. It drops half the features automatically

c. It sets all correlated weights to zero

d. It eliminates the bias term

Adding $\lambda > 0$ along the diagonal shifts all eigenvalues up by λ, making the matrix well-conditioned.

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40 [Challenging] • Target: Goodness of Fit and Regularization • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

If $R^2 = 0.64$ in simple linear regression, what is the magnitude of the Pearson correlation coefficient $|r|$?

a. 0.80 ✓

b. 0.64

c. 0.32

d. 0.40

In simple bivariate regression, $R^2 = r^2 \Rightarrow |r| = \sqrt{0.64} = 0.80$.

41 [Challenging] • Target: Linear Regression & OLS Practice Exercises • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

In a multiple linear regression model, the Sum of Squared Errors is $\text{SSE} = 100$ and the Total Sum of Squares is $\text{SST} = 500$. What is the coefficient of determination R^2?

a. $R^2 = 0.80$ ✓

b. $R^2 = 0.20$

c. $R^2 = 0.90$

d. $R^2 = 0.25$

The coefficient of determination measures the proportion of variance explained:
$R^2 = 1 - \frac{\text{SSE}}{\text{SST}} = 1 - \frac{100}{500} = 1 - 0.20 = 0.80$.

42 [Challenging] • Target: Simple Linear Regression and Residuals • Cross-Lesson: 7.01 Logistic Regression and Classification

In a multiple linear regression model, the Sum of Squared Errors is $\text{SSE} = 120$ and the Total Sum of Squares is $\text{SST} = 600$. What is the coefficient of determination R^2?

a. $R^2 = 0.80$ ✓

b. $R^2 = 0.20$

c. $R^2 = 0.90$

d. $R^2 = 0.25$

The coefficient of determination measures the proportion of variance explained:
$R^2 = 1 - \frac{\text{SSE}}{\text{SST}} = 1 - \frac{120}{600} = 1 - 0.20 = 0.80$.

43 [Challenging] • Target: Analytical OLS Formulas and Covariance • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

In a multiple linear regression model, the Sum of Squared Errors is $\text{SSE} = 20$ and the Total Sum of Squares is $\text{SST} = 100$. What is the coefficient of determination R^2?

a. $R^2 = 0.80$ ✓

b. $R^2 = 0.20$

c. $R^2 = 0.90$

d. $R^2 = 0.25$

The coefficient of determination measures the proportion of variance explained:
$R^2 = 1 - \frac{\text{SSE}}{\text{SST}} = 1 - \frac{20}{100} = 1 - 0.20 = 0.80$.

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44 [Challenging] • Target: Matrix Normal Equations and Projections • Cross-Lesson: 1.01 Vectors and Embeddings in AI

In a multiple linear regression model, the Sum of Squared Errors is $\text{SSE} = 40$ and the Total Sum of Squares is $\text{SST} = 200$. What is the coefficient of determination R^2?

a. $R^2 = 0.80$ ✓

b. $R^2 = 0.20$

c. $R^2 = 0.90$

d. $R^2 = 0.25$

The coefficient of determination measures the proportion of variance explained: $R^2 = 1 - \frac{\text{SSE}}{\text{SST}} = 1 - \frac{40}{200} = 1 - 0.20 = 0.80$.

45 [Challenging] • Target: Goodness of Fit and Regularization • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

In a multiple linear regression model, the Sum of Squared Errors is $\text{SSE} = 60$ and the Total Sum of Squares is $\text{SST} = 300$. What is the coefficient of determination R^2?

a. $R^2 = 0.80$ ✓

b. $R^2 = 0.20$

c. $R^2 = 0.90$

d. $R^2 = 0.25$

The coefficient of determination measures the proportion of variance explained: $R^2 = 1 - \frac{\text{SSE}}{\text{SST}} = 1 - \frac{60}{300} = 1 - 0.20 = 0.80$.

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